

Lec 52 Stable Diffusion Model

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Transcript

[00:00] Hello everyone. Welcome back to the lecture series on fundamentals of generative AI and large language models. In my previous session, we have discussed about the classifier free guidance in diffusion models. In today's session, we will talk about stable diffusion, which is based on latent diffusion model. Okay. So, the contents for today's session goes like the motivation, introduction to latent diffusion, latent diffusion training pipeline, latent diffusion objective function, stable diffusion pipeline, cross attention architecture, conditioning of stable diffusion model, advantages of latent diffusion model over DDPMs. Let us start with the motivation. What is the motivation for latent diffusion models and how it is

[00:48] related to stable diffusion model? Okay. So, in DDPM, we know that we take the entire image, which is a high dimensional image, and we perform the forward and the reverse diffusion in that high dimensional image. So, that is what is known as, you know, doing the forward and reverse diffusion on directly high dimensional pixel space images. Okay. So, every pixel is involved in my forward and reverse noising process. Okay. So, if X is my input image, it belongs to $\mathbb{R}^{H \times W \times 3}$, where H and W are the height and width of the image,

and 3 comes because it is an RGB image. Okay. So, for example, a 512 cross 512 cross 3, that means I'm taking an RGB image because 3 comes from

[01:47] there, contains 7864 432 pixel values. Okay. It's a huge number. These must be processed during every denoising steps making the process very computation- computationally very expensive. Okay. So, if you have to do the denoising process for each and every pixel, which is this number of pixels are there in the image, then the process becomes very, very expensive computationally. Okay. So, the problems that are associated is high memory consumption, computational cost is very high, and slow inference due to hundreds of denoising steps. Because we know in diffusion models, there are hundreds of denoising steps which are required to bring the image from a noise to the clean image. So, if you have to

[02:42] do the denoising steps in 786432, this a huge number of pixels, then we can understand how much time it will take for the inference, okay, or the image generation to happen, okay. So, from this motivation, what scientists thought is, can we compress the information of the image into something called as a latent space, okay. So, the core idea is compress the image into a compact latent representation and perform the diffusion in the latent state. We all know that in the latent space, the important information that is there in the image gets in a compressed It is It stays there.

[03:28] And what are the other irrelevant information are removed. So, if we take the compressed representation or latent space representation of the image and perform the diffusion steps on the latent space, it will reduce the computational cost of the model, okay. So, it was introduced by R. Rombach et al. in the paper high-resolution image synthesis with latent diffusion models and it came in the conference conference on computer vision and pattern recognition in 2022. Okay. So, this is a very significant contribution and we will see how this latent diffusion is related to stable diffusion. So, latent diffusion models are the algorithms that are introduced in this paper. So, the algorithm is latent diffusion and stable diffusion is

[04:22] a practical implementation of this LDM for text-to-image generation. So, there is a conditioning that is happening in stable diffusion model which works with algorithm of latent diffusion model. Okay. And in this case as we are compressing the image into the latent space representation the diffusion occurs in a 64 cross 64 uh cross uh four latent space four 64 cross 64, you know, latent space for a 512 cross 512 image. Okay.

[05:01] So, the diffusion occurs in the latent representation. Okay. Now, we will see what is there in the training pipeline of this latent diffusion. So, we have an input image X which goes to an autoencoder. The reason for putting it through the autoencoder is to compress it and Z is my latent representation which is E of X where E stands for the autoencoder or the encoder that encodes the information of the image into the latent space or latent representation. Now, this autoencoder is trained before

the diffusion and it is frozen. So, we will train the autoencoder, freeze it, keep it, and while training the diffusion model, we will use this frozen auto encoder. So, there is no training that is happening to the auto encoder while

[05:57] you are training the diffusion model. This auto encoder gives you the latent Z , which is your latent representation of your image. And this latent Z goes through forward diffusion process. And we all know the forward diffusion process gives us a noisy image from a clean image. So, here also the forward diffusion process gives you a noisy latent. And this noisy latent is the latent space equivalent of a noisy image in a DDPM. So, as simple as that. So, noisy latent is the latent space equivalent of the noisy image in DDPM.

[06:37] And then this noisy latent goes into the U-Net. We all know the U-Net is used for denoising and predicting the noise during the denoising or the reverse diffusion process. So, it undergoes reverse diffusion and it predicts the noise here, and we get a clean latent here, and then it goes through the decoder, which decodes it. So, D stands for the decoder. Z is your clean latent that is coming out of the predicted noise coming out of the U-Net, and $\hat{X}$ is your decoded uh you know, information. And then it is used for generating the image. Okay. So, this is the training pipeline for latent diffusion model. Okay.

[07:28] Now, what is the latent diffusion objective function? It is very similar to your DDPM objective function, where we have the predicted noise minus the uh actual Gaussian noises a difference, the square of the difference. So, in this case, instead of predicting noise in the pixel space, the network predicts noise in the latent space. So, we have seen the entire thing is happening in the latent space, so that is captured in our training loss, where L is the loss, E is the expectation value, and this is the, you know, $\epsilon - \epsilon_\theta$, which is ϵ_θ is our predicted noise.

[08:09] The prediction is happening on Z_t and t . Okay, Z_t is is the latent or the noise in the latent space, and ϵ is our Gaussian noise, actual noise that is added, and then you take the square of that one. So, Z_t is the noisy latent, ϵ is the Gaussian noise that has been added during the forward process. ϵ_θ is the predicted noise, and which we have to, you know, the important part is it it depends on Z_t and t , not on X_t . Okay.

[08:43] So, this is same as the DDPM objective, but applied to the latent vectors. Okay. So, that is the thing. Then we will see what happens in the stable diffusion pipeline. Okay. We have understood what is the concept of latent diffusion. Now we will look into how the stable diffusion pipeline like looks like because stable diffusion is used for text to image generation. So, there is a text prompt, which is used for image generation, and stable diffusion does it using latent diffusion model together with the text conditioning.

[09:22] Okay. So, we will see what exactly happen or how it happens. So, we have a

text prompt here, which is given as y . Y is my text prompt, say here. That text prompt goes through a text encoder, which is basically a pre-trained CLIP text encoder to understand the semantic information or the semantic meaning of the prompt. So, the text encoder's idea is to use a pre-trained encoder for understanding the semantic information that is present in the text prompt.

[10:02] After the text encoder, it goes for text embeddings. We all know that our models will not understand the text that is present. That it cannot read a text. We have to give the information in some form of mathematics in some mathematical form. So, which is done in the text embeddings. We uh know that text embeddings are given in the vector space and we call these embeddings are C equal to say this is $\tau_\theta Y$.

[10:38] And these embeddings represent the semantic meaning of the text in some numerical format. Okay? So, in some through some numbers. Or through some vectors. Okay? So, these text embeddings will be used for conditioning the generation image generation. Now, these text embeddings go through a layer of cross attention or go through a layer of or a block of cross attention. Okay? We will see how this part, which is your text conditioning part, which starts with the text prompt, how it is incorporated into the image generation part. Okay? Which is coming from our latent diffusion model.

[11:32] So, in our latent diffusion model, we have seen there is a noisy latent. What is that noisy latent? In my previous slide, I told noisy latent is nothing but the equivalent of noisy image in a DDPM, but in the latent space. Okay? So, that is the difference between noisy image that we tell in DDPM, here we call it as a noisy latent. Okay? And then, if that is Z_T , it goes to the unit. And this cross-attention is connected with the unit. And this is where the text enters.

[12:18] This is where the text information or text conditioning enters into the LDM algorithm. Okay? So, then the unit we know, unit predicts the noise, and we get the clean latent, and it goes to the decoder. Decoder also, we use the frozen decoder, like the encoder as I was telling. Encoder and decoder, they are trained before, and then they are used. And then, that decoded information is used for image generation. Okay? So, the important part to be uh you know, noticed here is there is a cross-attention block, which is a bridge between your text prompt and your latent diffusion model.

[13:14] And that helps in incorporating the text information or conditioning while generating the image using latent diffusion model. Okay? So, that is something which is which we have to remember. Now, we will see what does this cross attention do, why it is used, what will happen if we don't use the cross attention. So, as we all know, attention, whenever we speak about attention, there are three terms that comes into our mind, which is the query, key, and value. Okay? So, query, key, and value,

these are the three uh things that we compute while we are talking about attention mechanism.

[14:05] So, we all know query vector Q equal to $W_Q \phi I Z_T$. So, what exactly is this equation? Q is the query vector. Query is computed from the intermittent unit feature maps. That means it is coming from the image side. W_Q is the learnable weight matrix. Okay? $\phi I Z_T$, it is a feature map produced by the i -th layer of the unit. Okay? And here, $\phi I Z_T$ belongs to uh N cross D dimensional space, which denotes a flattened intermediate representation of the unit.

[14:57] Okay? So, the main point that we have to understand here is the query vector is coming from the image. Okay? It is coming from the feature maps from the unit. Okay? It is not coming from the text part. Okay? So, we have got the query vector. Okay? And then, we will see how we are getting the key vector. So, key vector K is equal to $W_K I$ and $\tau \theta I$. What is this $\tau \theta I$? If you remember, these were the text embeddings.

[15:41] W_K is the learnable weight matrix. And I is the I is basically the I -th layer. Okay? Of the unit. So, I -th layer that we are speaking about and that is where your text conditioning is going. So, what happens is the query comes from the image. Key comes from the text embedding. Okay? And value V value vector V is once again W_V into $\tau \theta Y$. That means W_V as we know, these are the learnable weight matrices or learnable weight matrix.

[16:24] And $\tau \theta Y$ is the text embedding. Okay? So, the query is coming from image part. Key and value are coming from the text prompts. That is the reason it is called as cross attention. Your query, key, and value are coming from different sequences. It is If it is coming from the same sequence, it becomes self-attention. But because your query comes from image and key and value are coming from your text embeddings, it becomes cross attention. Okay?

[17:06] Now, we all know cross attention weight equation. It's a known equation. So, attention QKV is given by the soft max $Q K^T$ by square root of D into V . Okay? So, that is where square root of D is the dimension. Okay? It's the dimension. And it is basically a scalar. Okay? And the next question is why cross attention is required. Okay? Suppose you have a text prompt like a yellow butterfly sitting on a purple flower. That is the text prompt you are giving for image generation.

[17:55] Okay? And you want that text information to be incorporated in your image properly. So, different images different regions of the latent image needs different words. So, we'll have we'll have a look into how it is. So, in the image you have butterfly, you have a flower, and a background. Now, your text prompt is giving a yellow butterfly. So, that yellow butterfly this information needs to go to the butterfly portion of the image.

[18:36] Okay? The In the latent region, whatever wherever is the butterfly, that needs to understand it should be a butterfly, and it should be yellow in color. Similarly, the latent region where the flower is there, it's needs the information that it is a flower, and it is purple in color. Okay? So, unless and until we give or we attend to these minute details that are given in the text prompt, those things will not come into the picture which is generated.

[19:15] And for the background, the remaining context, okay? So, basically, each and every part of your image requires attention from the text input. If it is a yellow butterfly, the text needs to give or the model needs to give attention to the butterfly and make it yellow. Okay? If the flower is purple, then the model needs to attend to that one and tell in the latent space, wherever the flower is there, wherever wherever the information about the flower is there, that needs to be shaped like a flower, first of all, and secondly, the color should be purple. Okay? So, this is the reason why we require the cross-attention.

[20:02] Now, we will look into the architecture. How the entire architecture of this stable diffusion looks like or which is based on your latent diffusion model, okay? Now, so, this is an image that I have taken from the original paper, and we will look into the different blocks. There are three blocks here, the pink-colored one, the green-colored one, and there is another white one. Okay? So, we will look into how this latent diffusion model is explained through this architecture and how conditioning is happening for stable diffusion, how the entire thing is coming into picture. So, we will have a look into that one. So, first, we will we will study it block by block. The first one is my the pink block,

[20:55] which is called as the pixel space, and we will see what happens in the pixel space. There are two things that is happening here. The original image is X . Then, it is going through the encoder E . Okay, so that encoder gives me the Z , which is basically my latent representation of the original image X . Okay. So, this is happening in this part. What is happening here? The decoder D reconstructs the image. So, this latent Z goes through my entire path here.

[21:36] Okay. And then something something happens here. Again, it comes back. We will see into what are those something something in my next slide. And then what I see here is the decoder is working on Z . And then we get the reconstructed image, which is my $\hat{X}$, okay, or $\tilde{X}$. So, this is what is present in my pixel space. Now, we will see what is there in the latent space. So, once again, the picture is here in front of us, and we have this latent space into the discussion.

[22:18] So, what exactly is happening in the latent space? We will see. So, in the latent space, the entire diffusion process happens there. Forward, reverse, all diffusion pro- process happens in the latent space. Gaussian noise is progressively added to the latent space representation. So, if you see, this is my latent representation of my

image that has come from the pixel space, and then it is going through the forward diffusion process where uh noise is added progressively, and we get Z_t at time instant small t equal to capital T .

[23:01] Okay? » [snorts] » And the unit Now, here you have the denoising unit, which predicts epsilon theta, which is your predicted noise. And that prediction happens in the reverse denoising process. So, the Z_T goes through this one. And in the unit, you can see there are These are the skip connections that are happening here. The skip connections, as we know, that they are used to keep the finer details in the reconstructed image, which may have been lost during the encoding part.

[23:43] And the skip connections help us to put those finer details back into the image in the decoder part. Okay? So, the unit learns to remove this noise during the reverse diffusion that we know. And if we see query, key, and value, there are four blocks of cross attention that are placed at or applied at several resolutions. So, this is one of the main or the main innovation of this paper. Okay? So, the cross attentions are placed everywhere in the denoising everywhere in the latent space.

[24:22] And to be marked is it is placed at several resolutions. Okay? This helps in generating the image or image details with all the conditioning that is coming from the text input or other input that is coming as a conditional information, conditioning information. Okay? Now, we will look into So, we have seen the pixel space, we have seen the latent space, they're both are done. Then, we will look into this block, which is the conditioning block.

[24:58] What is happening there? There are a lot of things that are written here. Semantic map, text representation, images, and then tau theta. What is this? Okay, so we will have a look here. So, the stable diffusion supports multiple conditioning modalities, okay? So, the primary one for which, you know, the stable diffusion was used was text prompts for text prompts to image generation or image generation using text prompts, but it can be used for other semantic other modalities also, which is like semantic maps or image to image generation also.

[25:43] And that conditioning information comes here. Okay? So, we will see how the conditioning happens or what exactly I mean by telling by conditioning. So, if it is a semantic map I have, so there is a dog, cat, two cats are there. So, this is a semantic map and I want it to change into an image. So, that is called semantic map to image generation. Then, we have text prompting, which is like cat sitting on a red carpet and you have an image where a cat actually is sitting on a red carpet.

[26:17] And image to image translation, say from black and white to colored image. So, these are the various conditioning mechanisms or conditioning modalities in which we can use the stable diffusion. And that is where what our conditioning block is showing. This [clears throat] is shown here, okay? Now, we will look into there are a couple of things, denoising steps, cross attention, switch, skip connection, and

concatenation.

[26:48] These are present in my image. So, we understand the denoising we understand the cross attention, we understand the skip connection. What is this concatenation? From where it is coming? And what is this switch? What exactly it is doing here? What is the reason for putting the switch here? So, we will have a look there. So, we'll have first have a look into the concatenation. So, if we see this conditioning in stable diffusion, as I was telling, it can be for various modalities.

[27:23] It can be a text prompt, it can be, you know, semantic map, it can be another image. Okay? So, that conditioning can happen in or the conditioning information can be put in the architecture using two methods. One is concatenation and the other one is through cross attention. So, we will see when to use concatenation. Concatenation is used when conditioning has spatial information. So, when do we expect spatial information?

[28:00] Will my text prompt have a spatial information? No. Spatial information is possible when we have a segmentation map, when we have a image, it can be a low resolution image, or any other image for that matter. So, in those cases, wherever spatial information is present, we use concatenation for adding that information in my diffusion model, okay? And we use cross attention for text embeddings, where no spatial dimensions or spatial information is there, okay?

[28:40] And what is this switch? What exactly we mean by switch? So, depending on the application. Either we go for or the model chooses concatenation or cross attention. So, depending on the application, we use either concatenation or cross attention. So, that is what is you know, implied by putting a switch. So, depending on the application, if you are doing a text prompt to image generation, we will use cross attention. If we are using image to image generation, we will use concatenation.

[29:23] Okay? So, that is the reason for writing that switch. Okay? Now, we will look into the advantages of stable diffusion over DDPM. I have already spoken, but just to you know, uh give uh little more importance. So, DDPM adds Gaussian noise to a high-resolution photograph until it becomes pure noise. We are reinforcing that one. And uh a stable diffusion in stable diffusion, what happens? First, we compress the photograph into something compact, which is the latent space.

[29:58] And we call it we are calling it as a blueprint because it is having all the important information that is required for further processing. We have removed all the unimportant information from there. Then, we have added Gaussian noise to that blueprint. Okay? And during generation, the model cleans this noisy blueprint step by step and finally reconstructs the full image of that one. Okay? So, the entire noising and denoising process, as we have discussed before also, uh that it happens in the latent space.

[30:30] In the latent uh representation of the image. This makes stable diffusion much more computationally efficient because here the number of pixels on which your uh uh noising or denoising will work, denoising particularly when you are in the inference stage, that number is reduced here, which reduces the computational cost. Okay. So, that is the main difference which we have spoken before. In this slide, we have reinforced it once more.

[31:02] And with this one, I come to almost the end. Here is an image which I have taken from the original paper, text-to-image synthesis, and uh on Leon uh 1.45B model. So, you can see there are two different rows, and these are the prompts that have been given, a street sign that reads latent diffusion. So, there are two images that have been shown here with the same text from text prompt. That means the same text uh for uh prompt can give you two variations of the image. So, a zombie in the style of Picasso.

[31:48] So, there are two zombies which are generated in two separate runs, and image of an animal half mouse, half octopus, and then illustration of slightly conscious neural network. Okay, and so on. So, these are the images that are taken from the paper to express how text-to-image synthesis happened. And there are two variations that have been generated from the model has been shown. There can be multiple variations that can be generated from the model.

[32:24] So, with this one, I come to the end of today's session. Let Before concluding, let me summarize what we have studied today. So, we have studied the latent diffusion model, which is the back which is uh latent diffusion model is the backbone for any stable diffusion model. And how it is different from normal DDPM? Here, the concept is very simple. Here, we do not do any noising denoising things in the entire image. We compress it into a latent space and that latent space in latent space representation of the image is taken through the diffusion steps forward as well as reverse and that helps in minimizing the computational cost that is required. Okay. So, that is the main core idea of you know, this latent diffusion models.

[33:18] And then we have seen in stable diffusion model, latent diffusion model is the algorithm that is introduced in the paper and stable diffusion model is a practical implementation of that one with text prompt conditioning. So, we have looked into how these text prompt conditioning is incorporated in the latent diffusion model through attention cross attention blocks. » [clears throat] » The cross attention blocks help to give all the information to the image to various parts of the image and get the image generated with all the finer details that are coming from the prompt. Okay. So, that is the reason for the cross attention and then we have seen into the pipeline, the stable diffusion pipeline, the latent diffusion pipeline.

[34:08] Then we have seen into the architecture, the pixel space, the latent space and the conditioning block. What exactly happens in each of these blocks? Then we

have seen that there are two methods by which you know, the text information can or or the conditioning information can go into the model. One is concatenation and another is uh, you know, cross attention. Cross attention is used when there is no special information that is present in the conditioning information.

[34:39] Where Whereas anywhere the special information is present or special uh, yes, information is present for conditioning, there we use concatenation. So, and it is depending on the application, either concatenation or cross attention has is is chosen. And then we have seen what are the advantages. We already know the advantages. It reduces the computational cost of uh, the model by compressing the image into the latent space and then doing the diffusion process. Okay.

[35:15] So, with this one, I come to the end of uh, today's uh, session. In my next session, we will go to a completely uh, new topic which is the foundations of natural language processing. And from there, we'll go forward towards transformers and large language models. And so, today's session is the last session where we have talked about image generation using diffusion models. Okay. So, I urge all of you to go through the original paper that I have shared here in the slide.

[35:48] And uh, till I mix meet all of you in the next session, please keep learning, happy learning, and thank you all.